

COURSE SYLLABUS

CSC15006 – Introduction to Natural Language Processing

1. GENERAL INFORMATION

Course name:	Introduction to Natural Language Processing
Course name (in Vietnamese):	Nhập môn xử lý ngôn ngữ tự nhiên
Course ID:	CSC15006
Knowledge block:	Fundamental and specialized knowledge
Number of credits:	4
Credit hours for theory:	45
Credit hours for practice:	30
Credit hours for self-study:	90
Prerequisite:	
Prior-course:	
Instructors:	Dinh Dien, Nguyen Hong Buu Long

2. COURSE DESCRIPTION

This course introduces the fundamental concepts, methods, and applications of Natural Language Processing (NLP), a core area of Artificial Intelligence that enables computers to process and understand human language. Students will explore linguistic foundations, text preprocessing, statistical and neural language models, word representations, and parsing techniques, progressing toward advanced approaches with sequence-to-sequence architectures and transformers. Practical applications such as text classification, sentiment analysis, machine translation, summarization, and conversational AI will be emphasized, along with discussions on current trends, challenges, and ethical considerations in NLP. Through lectures, hands-on exercises, and projects, students will gain both theoretical knowledge and practical skills to build and evaluate NLP systems.

3. COURSE GOALS

At the end of the course, students are able to

ID	Description	Program LOs
G1	Develop a solid understanding of corpus-based approaches in NLP and apply them to core tasks such as part-of-speech tagging, sentence parsing using statistical methods, word sense disambiguation, information retrieval, text classification, and statistical machine translation...	LO1, LO2
G2	Understand theoretical knowledge of linguistic aspects including morphology, grammar, and semantics, and understand their importance in computational processing of natural language.	LO6, LO7
G3	Apply morphological, grammatical, and semantic analysis techniques to real-world text processing problems, with a particular emphasis on handling Vietnamese text alongside English.	LO6, LO7
G4	Evaluate and compare the performance of NLP systems, using standard metrics and methodologies to assess effectiveness, accuracy, and limitations.	LO3, LO4, LO5, LO7
G5	Strengthen academic and professional skills such as reading and analyzing English-language technical materials, collaborating effectively in teams, and preparing and presenting project reports.	LO12

4. COURSE OUTCOMES

CO	Description	I/T/U
G1.1	Know and understand specialized English terminology of the subject.	I

G1.2	Ability to apply English language processing methods for Vietnamese.	I
G1.3	Classifying basic concepts and terms in natural language	T
G1.4	List the similarities and differences between English and Vietnamese in terms of automatic language processing	T
G2.1	Understand some problems in English and Vietnamese natural language processing.	TU
G2.2	Applying knowledge of Natural language processing to deploy application problems	IU
G3	Using a number of programming languages, organizing appropriate data structures, understanding and implementing available algorithms.	I
G4.1	Ability to self-learn other methods (in addition to the learned content) for Natural Language Processing problems.	I
G4.2	Applying teamwork skills	U
G4.3	Applying skills to search, read and understand professional documents, and write reports	U
G5.1	Read and understand English materials related to lectures.	U
G5.2	Written and oral skills, and presentation skills related to the subject's topics.	U

5. TEACHING PLAN

THEORY

ID	Topic	Course outcomes	Teaching/Learning Activities (samples)	Assessments
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1	Introduction to natural language processing (NLP): - Overview of NLP; - NLP applications; - Tools & Resources; - Challenges in NLP.	G1.1,G1.2, G1.3,G1.4 G2.1	Lecturing Q&A, Group discussion	
2	Linguistic Fundamentals for NLP: - Phonetics and Phonology; - Morphology; - Syntax; - Semantics; - Pragmatics; - Discourse & Context.	G1.1,G1.3, G2.1	Lecturing Q&A, Group discussion	
3	Text Preprocessing & Representation: - Text Normalization; - Tokenization; - Stopword Handling; - Stemming & Lemmatization; - Vector Representations of Text.	G1.1,G1.2, G1.3,G1.4, G2.1	Lecturing Q&A, Group discussion	
4	Language Models: - Introduction to Language Modeling; - n-gram Language Models; - Data Sparsity Problem; - Smoothing Techniques; - Evaluation Metrics; - Limitations of n-gram Models.	G1.2,G1.3, G1.4,G2.1, G2.2	Lecturing Q&A, Group discussion	

5	Syntax & Parsing: - Introduction to Syntax in NLP; - Part-of-Speech (POS) Tagging; - Constituency Parsing; - Dependency Parsing; - Syntactic Ambiguity & Challenges.	G1.2,G1.3, G1.4,G2.1, G2.2	Lecturing Q&A, discussion	Group	
6	Text Classification: - Introduction to Text Classification; - Feature Extraction for Classification; - Machine Learning Approaches; - Neural Network Approaches; - Evaluation Metrics.	G1.2,G1.3, G1.4,G2.1, G2.2	Lecturing Q&A, discussion	Group	
7	Sequence-to-Sequence Models: - Introduction to Seq2Seq; - Encoder–Decoder Architecture; - RNN-based Seq2Seq Models; - Attention Mechanism; - Decoding Strategies; - Beyond RNNs.	G1.2,G1.3, G1.4,G2.1, G2.2	Lecturing Q&A, discussion	Group	
8	Advanced NLP with Transformers: - Motivation for Transformers; - Transformer Architecture; - Pretrained Language Models; - Fine-tuning and Transfer Learning; - Efficiency & Scaling; - Challenges & Limitations.	G2.1,G2.2, G3,G4.1	Lecturing Q&A, discussion	Group	
9	Applications & Future of NLP: - Major Applications of NLP;	G2.2, G4.1,G4.2,	Lecturing Q&A,	Group	

	- Recent Trends & Advances; - Ethical & Societal Challenges; - The Future of NLP.	G4.3,G5.1	discussion	
10	Coursework report	G3, G4.1, G4.2,G4.3, G5.1,G5.2	Demonstrations, Debates	Project

LABORATORY

ID	Topic	Course outcomes	Teaching/Learning Activities (samples)	Assessments
1	Automatic OCR with PPOCRLabel and write comments on the results	G2, G3, G4, G4, G5	Group discussion	LW#1
2	Detect and correct OCR errors from bounding box and character results	G2, G3, G4, G4, G5	Group discussion	LW#2
3	Automatic OCR processing on multi-page files	G2, G3, G4, G4, G5	Group discussion	LW#3

6. ASSESSMENTS

ID	Topic	Description	Course outcomes	Ratio (%)
A1	Assignments			20%
A11	Labwork: LW1–LW3	LW1: Basic OCR methods LW2: Detect and correct OCR errors LW3: OCR processing on multi-page files	G2, G3, G4, G4, G5	20%
A2	Mid-term Projects			40%

A21	Project	Students (individually) will use programs or tools (from the Labwork section) to automatically build a parallel corpus or “monolingual corpus” or “dictionary/dictionary from raw corpus sources” that will be provided to each student.	G2, G3, G4, G4, G5	40%
A3	Final Projects			40%
A31	Theory final exam	Groups of students (3-4 students) will choose a project according to the teacher's regulations. The group of students will train their model based on the corpus that their classmates have built in their midterm projects.	G1.1,G1.2, G1.3,G1.4 G2.1,G4.1, G4.3,G5.1, G5.2	40%

7. RESOURCES

Textbooks

- Daniel Jurafsky, James H. Martin. “Speech and Language Processing: An Introduction to Natural Language Processing, Computational Linguistics, and Speech Recognition”, 2025 (3rd ed. Draft).
- Hobson Lane and Maria Dyschel. “Natural Language Processing in Action”, January 2025, ISBN 9781617299445, 688 pages.
- Husn Ara, “Mastering Natural Language Processing: Beginner to Advanced Level Guide”, October 18, 2024, ISBN 979-8343611366, 476 pages.
- Andrew Davis, “Natural Language Processing with Artificial Intelligence: Building Smarter Applications Through Language Understanding”, October 8, 2024, 140 pages.

- Đinh Điền. “Xử Lý Ngôn Ngữ Tự Nhiên”. NXB ĐHQG, 2006

Others

- Christopher D. Manning, Hinrich Schütze. “Foundation of Statistical Natural Language Processing”. MIT Press, 2001.
- Vaclav Brezina. “Statistics for Corpus Linguistics”. Cambridge University Press, 2018.
- <http://nltk.sourceforge.net/>

8. GENERAL REGULATIONS & POLICIES

- All students are responsible for reading and following strictly the regulations and policies of the school and university.
- Students who are absent for more than 3 theory sessions are not allowed to take the exams.
- For any kind of cheating and plagiarism, students will be graded 0 for the course. The incident is then submitted to the school and university for further review.
- Students are encouraged to form study groups to discuss on the topics. However, individual work must be done and submitted on your own.